



2022 Condition report

Bridges and Major Culverts



Uganda National Roads Authority

Uganda National Roads Authority
Network Planning Department

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1 INTRODUCTION

Bridges are an integral part of the road network and therefore require optimum management to preserve the road network. They are a key element with a high asset value and as such require efficient care to minimize losses. The failure of a bridge also poses a high safety risk to road users and highly disrupts traffic. As a result, attention must be given to the proper management of bridges and major culverts on the management.

The first step to a proper management system is regular inspection followed by timely maintenance. This is to ensure the safety of the road users and to protect the capital investment of the structure with a minimal operational cost.

This document reports on the condition of the bridges and major culverts which will inform the maintenance needs of the structures on the network.

2 UNRA STRUCTURES

UNRA classifies its structures as Bridges and Major Culverts.

2.1 Definition of a Bridge

A structure is classified as a bridge if one or more of the following is true:

- Any single span, measured horizontally at the soffit along the road centreline between the faces of its supports ≥ 6 m, or
- The individual clear spans > 1.5 m, and overall length between abutment faces > 20 m, or
- Opening height ≥ 6 , or
- Total cross-sectional opening $\geq 36\text{m}^2$, or
- Structure is a road-over-rail (even if span < 6 m)
- If the structure consists of clearly identifiable elements such as deck slabs, deck expansion joints, abutments, piers and foundation footings. Elements such as invert slabs, apron slabs, cut-off walls are normally not present. Should a structure with dimensions of a Major Culvert present mostly bridge elements, then it should be considered a bridge and recorded as such in the BMS.

Bridges are named as Bxxx with xxx denoting the unique number assigned to each bridge.

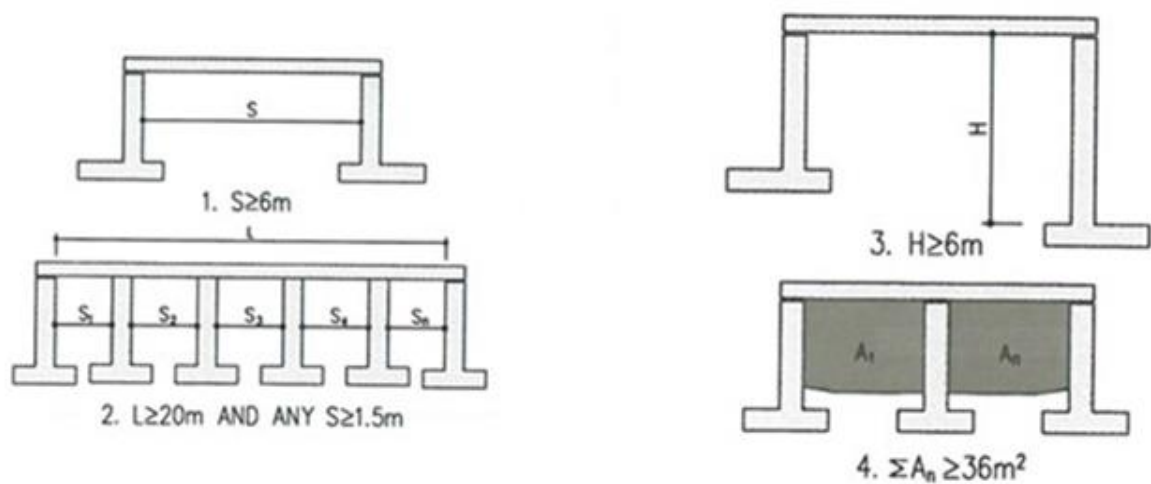


Figure 2-1: Definition of a Bridge

2.2 Definition of a Major Culvert

A Major Culvert is a structure, normally “buried” with road fill or road slabs on top, consisting of “cellular” units. A cellular unit can typically be described as an “opening” where, in general, the overall cell length is greater than the cell width. Elements such as separate deck slabs, abutments/piers, foundations, etc. are not clearly identifiable while elements such as invert slabs, apron slabs, cut-off walls etc. are normally present. Elements such as separate deck slabs, abutments/piers, foundations, etc are NOT clearly identifiable. Should a structure have dimensions exceeding that of a Major Culvert but presenting mostly culvert elements, then it should be considered a Major Culvert and recorded as such in the BMS.” Culverts with a pipe span of less than 2m are termed Minor Culverts and are not included in the BMS.

Major Culverts are named as Cxxx with xxx denoting the unique number assigned to major culvert.

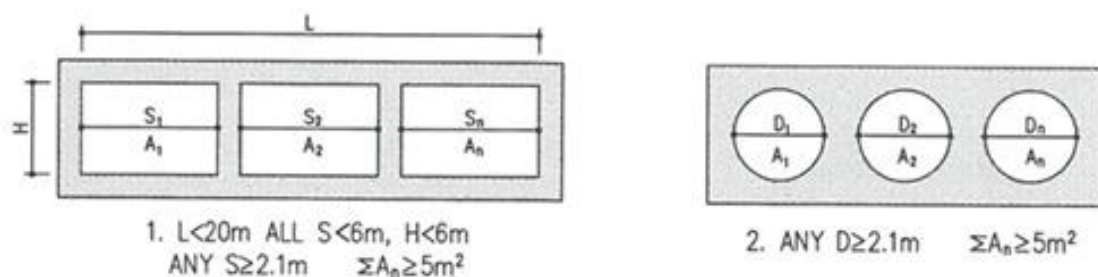


Figure 2-2: Definition of a Major Culvert

2.3 Structures on the UNRA network

Based on the assessment of FY 2021/22, the total number of structures on the network is 816. These are categorized as Bridges and Major Culverts as the two have different maintenance needs and life spans. For this cycle (FY2021/22) inventory and condition assessment of these structures was done for the Northern, Southern and Western regions. Historic data was used for the unassessed structures. Validation of this data has been ongoing, and the numbers are likely to change due to the construction of new structures and upgrading of roads. Inventory of some of the roads under upgrade has not yet been covered.

The graphs below detail the number of structures on the network.

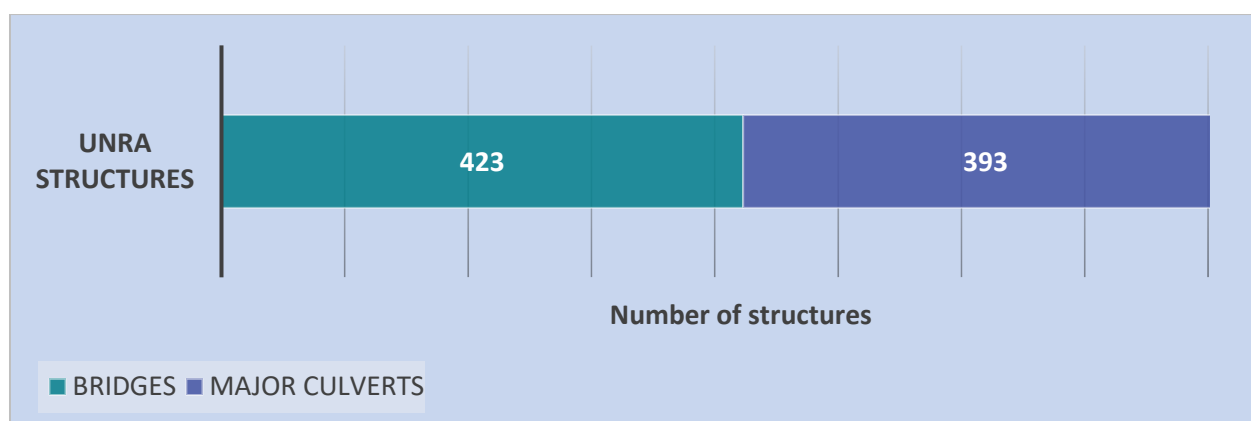


Figure 2-3: Split between Bridges and Major Culverts on the UNRA Network, 2022

From Figure 2-3, it is observed that the network has more Bridges than Major Culverts.

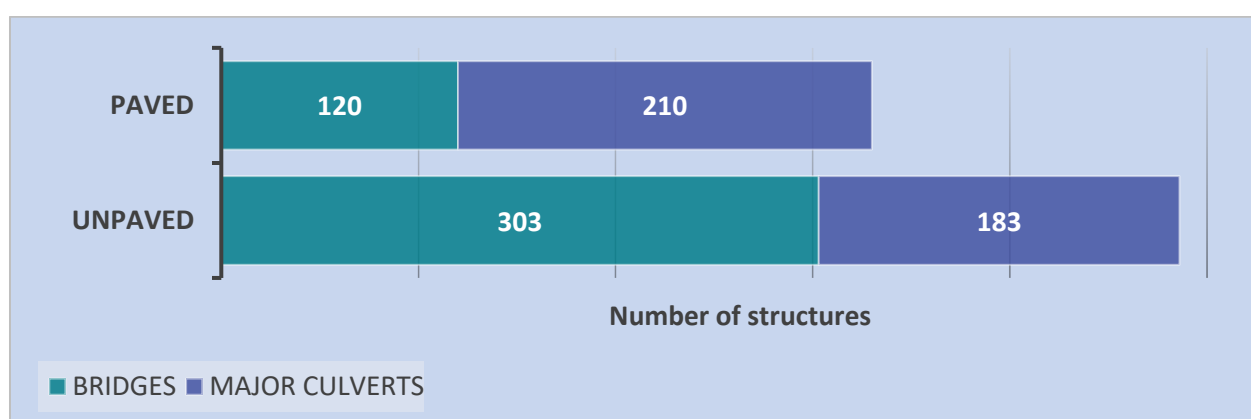


Figure 2-4: Split between Bridges and Major Culverts on the Paved and Unpaved Network, 2022

From Figure 2-4, it is observed that most structures are on unpaved roads.

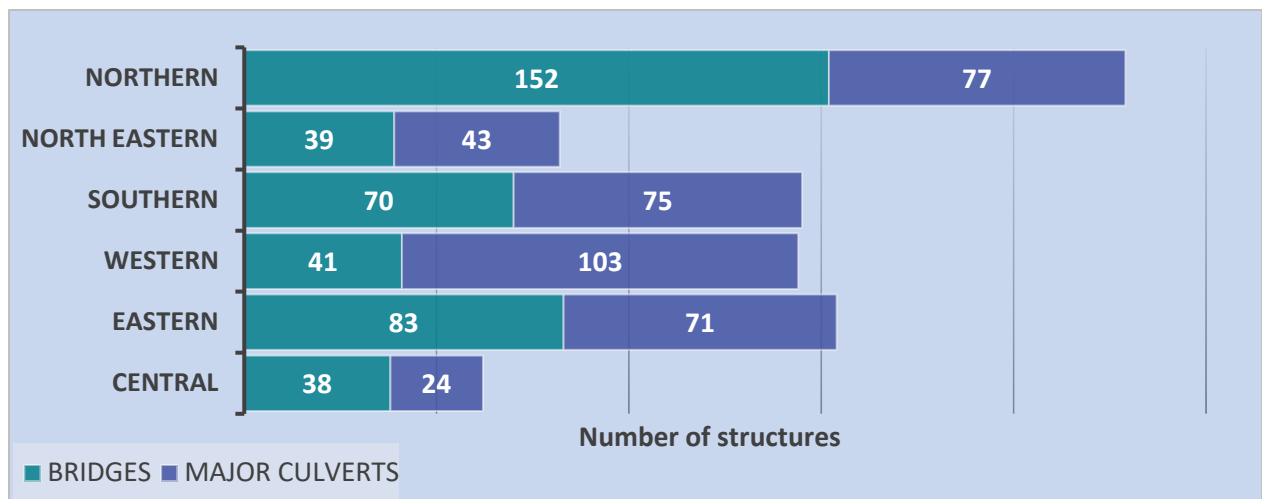


Figure 2-5: Split between Bridges and Major Culverts per region, 2022

From Figure 2-5 above, it can also be observed that the Northern region followed by the Eastern regions have the highest number of structures while the Central region has the smallest number of structures. This has an impact on the funding levels for routine maintenance of structures per region.

3 INSPECTIONS AND THE BMS

UNRA uses the Bridge Management System (BMS) to effectively manage and maintain its structures. The BMS provides an avenue where UNRA can:

- Comprehensively store all inspection data on its structures
- Report on the inventory and condition of structures
- Identify defects and required interventions
- Plan and prioritize maintenance and upgrade works on structures
- Estimate short- and long-term budgetary requirements to maintain its structures at the required level of service

To populate the BMS, visual inspections are carried out to ascertain the inventory of the structure, identify defects, the stage and extent of the defect and the maintenance recommendations. Inspections are carried out by qualified bridge inspectors and must consist of a thorough examination of all the components of the structure. For bridges, the components assessed are Waterways (if applicable), Foundations, Substructure (Abutments and Piers), Bearings, Superstructure, Non-structural elements (joints, drainage, parapets etc.) and Approaches. For Major Culverts, the components assessed are Waterways, Inlets and Outlets, Substructure and Superstructure (Cell

Invert, Cell Roof Slab, Cell walls), Precast Pipe Culvert, Corrugated Steel Pipe and Roadway.

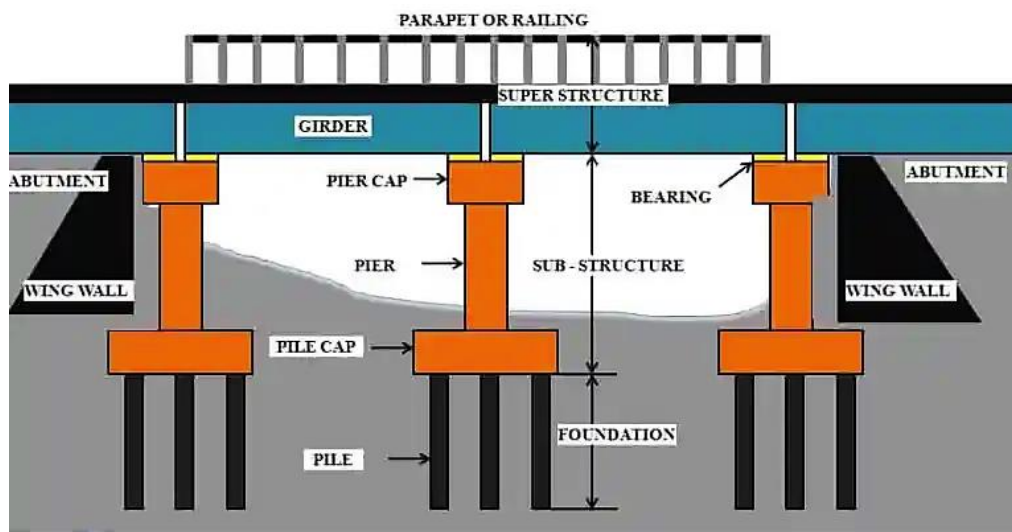


Figure 3-1: Components of a bridge

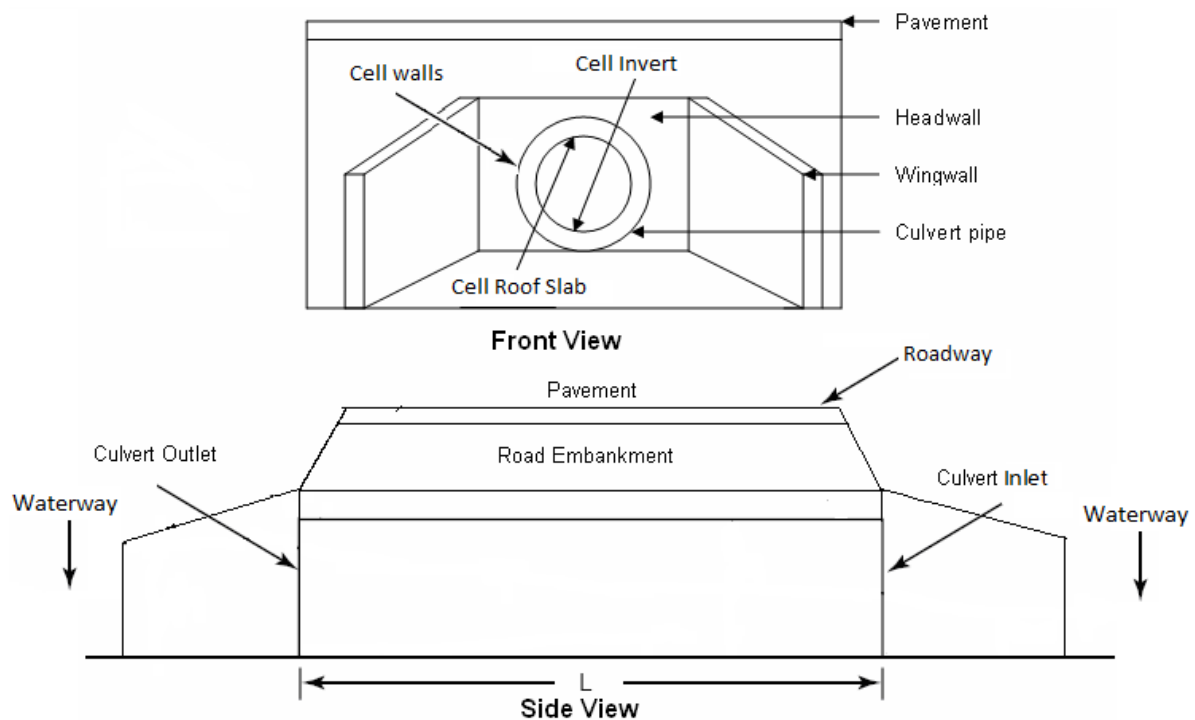


Figure 3-2: Components of Major Culverts

A numerical scale (0-9) is used to rate the condition of both the main components and subcomponents. These ratings are based on the current prevailing condition of the component and are used to calculate an overall rating of the structure. The Overall Rate calculated is classified in 5 condition categories, i.e. critical, poor, marginal, satisfactory, good, as follows;

Table 3-1: Condition categories of the Overall Rating

Calculated Overall Rate	Condition Category
0 and 1	Critical
2 and 3	Poor
4 and 5	Marginal
6	Satisfactory
7 to 9	Good

4 CONDITION OF BRIDGES AND MAJOR CULVERTS

4.1 Condition of Bridges

Inventory and condition assessment of bridges and major culverts was done by the Data Collection Consultants in the Northern, Southern and Western region for the year 2022. A delay in the contract signing of Lot 1 (Central region) and issuance of Call off order 2 for Lot 4 (Eastern/North-Eastern region) rendered it impossible to have a full network assessment by the time of this report. For purposes of this analysis, historical assessment data was used for the unassessed structures. Based on this, the following conclusions can be made with regards to the condition of Bridges in UNRA.

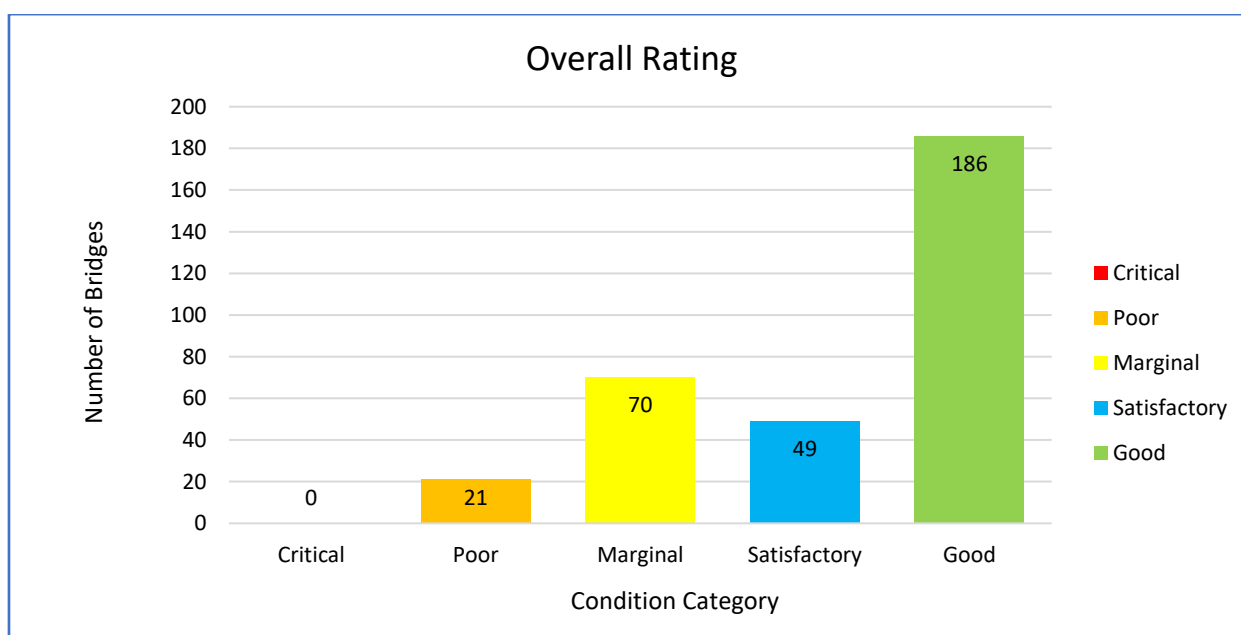


Figure 4-1: Overall rating of bridges

From Figure 3-3 above, it can be observed that the highest number of bridges is in good condition (186No.) and 49No. bridges are in satisfactory condition. The bridges in marginal condition are significant (70No.) and timely maintenance is advised to prevent them from quickly deteriorating to a poor condition. Bridges in poor condition

are the least (21No.) and urgent intervention is required. No bridge is in critical condition. Appendix 6.1 details the bridges in poor condition.

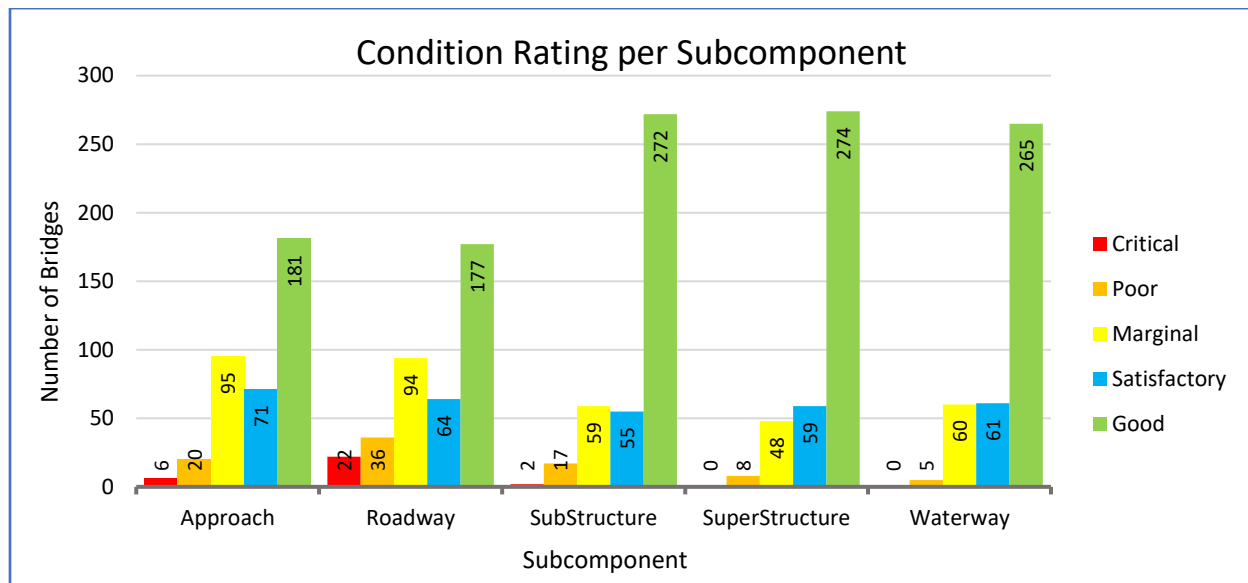


Figure 4-2: Condition rating of bridges per subcomponent

From the figure above, it is observed that;

- The majority of bridges have subcomponents in good condition.
- An appreciable number of subcomponents have a marginal rating and timely maintenance is required.
- Subcomponents in poor and critical condition are minimal and displaying in only the approaches, roadway and substructure and urgent maintenance is required. Failure to do so possess a safety hazard and can lead to closure or collapse of a bridge. See Appendix 6.1 for the bridges with components in critical and poor condition.

4.2 Condition of Major Culverts

Inventory and condition surveys were carried out as stated in 4.1 above. Based on these visual assessments, the following conclusions can be made with regards to the condition of Major Culverts in UNRA.

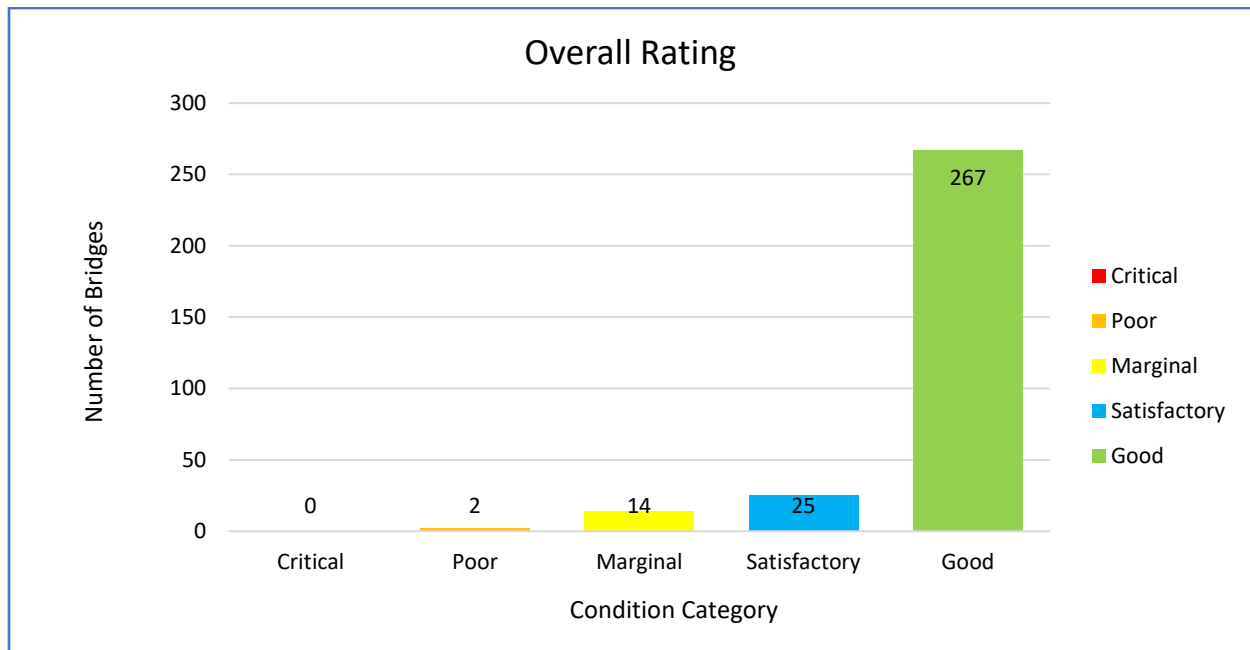


Figure 4-3: Overall rating of major culverts

From Figure 4-3 above, it can be observed that 267No. major culverts are in good condition. 25No. are satisfactory and 14No. are in marginal condition. Only 2No. major culverts are in poor and critical condition and immediate intervention is required (see Appendix 6.2 for details).

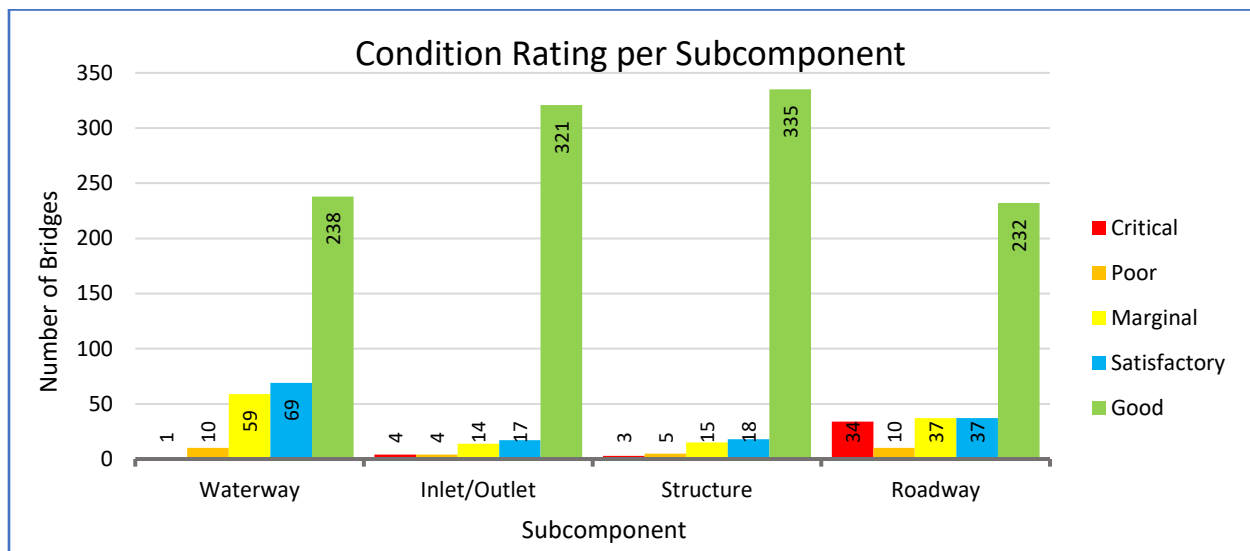


Figure 4-4: Condition rating of Major Culverts per subcomponent

From the figure above, it is observed that;

- The majority of major culverts have subcomponents in good condition.
- Subcomponents in poor and critical condition are minimal and displaying in mostly the roadway. More effort is required in timely maintenance of the

roadway. See Appendix 6.2 for the major culverts with components in critical and poor condition.

5 CONCLUSION

The UNRA network has 816 structures with 423 bridges (52%) and 393 Major Culverts (48%).

Based on the assessed structures, the condition of bridges is generally good with 56% (186No.) in good condition, 15% (50No.) satisfactory, 22% (73No.) marginal, 6% (21No.) poor and 0% (0No.) in critical condition. Considering the condition of the subcomponents, the roadway has the highest number of bridges in critical condition 6% (22No.).

The condition of major culverts is generally good with 85% (275No.) in good condition, 10% (31No.) satisfactory, 5% (15No.) marginal, 1% (2No.) poor and 0% (0No.) in critical condition. Considering the condition of the subcomponents, the roadway has the highest number of major culverts in critical condition 10% (34No.).

6 APPENDICES

6.1 Bridges in poor condition

S/N	Bridge No.	Bridge Name	Link ID	Link Name	Chainage (km)	Maintenance Station	Maintenance Region	Number of spans	Bridge Length	Bridge Width	Overall Rating
1.	B080	Mubuku	C510_Link01	Mubuku-Maliba-Nyakalingijo	10.7	Kasese	Southern	1	26	5	Poor
2.	B098	Nkola	C833_Link01	Nabumali junction-Busolwe	34.488	Tororo	Eastern	1	7	4	Poor
3.	B104	Sonoli	C862_Link01	Namagumba - Budadiri	18.427	Mbale	Eastern	1	7	8	Poor
4.	B105	Namatsu	C863_Link01	Mbale - Nkokonjeru	21.067	Mbale	Eastern	1	8	10	Poor
5.	B109	Tsutsu	C867_Link02	Bumbobi-Bubulo-Budud	28.48	Mbale	Eastern	1	9	8	Poor
6.	B150	Agora	B252_Link02	Lira-Acholibur-Kitgum	104.78	Kitgum	Northern	1	5	8	Poor
7.	B176	Lakapelot	B301_Link03	Kotido - Apaan - Kabong	102.411	Kotido	North Eastern	1	16	7	Poor
8.	B189	Awere	C680_Link02	Gulu-Opit-Rwackoko	74.91	Gulu	Northern	3	50	4	Poor
9.	B215	Lojoro	C960_Link01	Angatun-Nabilatuk-Lokapel	46.817	Moroto	North Eastern	1	6	4	Poor
10.	B323	Agagura	C682_Link01	Gulu - Logere - Adeo		Gulu	Northern	2	9	5	Poor
11.	B338	Acodho	C747_Link01	Panyimur -- Vurra Customs	18.264	Arua	Northern	1	7	4	Poor
12.	B348	Kochi	C758_Link01	Keriri - Midigo	6.7	Moyo	Northern	4	24	4	Poor
13.	B352	Korubude	C836_Link02	Mulanda - Nagongera		Tororo	Eastern	1	4	4	Poor
14.	B354	Kaminima	C837_Link01	Nagongera - Merikit	18.218	Tororo	Eastern	1	5	5	Poor
15.	B355	Kamunima	C837_Link01	Nagongera - Merikit	18.218	Tororo	Eastern	1	5	5	Poor
16.	B362	Nahamya	C873_Link01	Nabiganda - Kaiti	3.436	Mbale	Eastern	2	15	4	Poor
17.	B383	Omongo	C964_Link02	Toroma-Magoro	17.5	Soroti	North Eastern	2	12	4	Poor
18.	B864	Kiguli	C862_Link01	Namagumba - Budadiri	23	Mbale	Eastern	1	10	7	Poor
19.	B874	Bukwo	B304_Link02	Kaserem-Kapchorwa-Suam	98.69	Mbale	Eastern	1	21	6	Poor
20.	B879	Lonene	C968_Link01	Matany - Lokopo - Turtuko	29.827	Moroto	North Eastern	2	19	3	Poor
21.	B886	Kwakiteza	B306_Link01	Tororo (Malaba junction) - Busia	23	Tororo	Eastern	1	11	5	Poor

6.2 Major Culverts in poor condition

Culvert No	Link No	Link Name	Chainage (km)	Maintenance Station	Maintenance Region	Type of culvert	No. of pipes	Overall Rating	Diameter (m)
C011	C317_Link01	Katuugo-Kinyogogga	12	Luwero	Central	Corrugated metal pipe culvert	6	Poor	2.8
C558	C726_Link01	Adjumani - Sinyanya Ferry landing	14	Moyo	Northern	Concrete pipe culvert	3	Poor	1.8

6.3 Pictures of structures in poor and critical condition

6.4 List of Bridges on the Network

6.5 List of Major Culverts on the Network